Nikolas Dinh

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EDUCATION

California Polytechnic University, Pomona Expected Graduation Fall of 2027
Mechanical Engineering Student 3.5 GPA

SKILLS

CAD: Solidworks **(CSWP)**, AutoCAD, Fusion 360, OnShape,
Electronics/Testing: Arduino, Raspberry Pi, Soldering, Strain Gauges, Tensile Testing,
Software and Programming: MATLAB, Ansys Fluid Flow Fluent, Python, C++, Ansys Mechanical FEA (Spring 2027)

WORK EXPERIENCE

VSE Corporation — Manufacturing Engineering Intern June 2026 - August 2026

- Increased servicing capabilities for brake line conversions and older wheel flanges by modeling special tooling in **SolidWorks**, with proper **GD&T** for CNC machining allowing support for more aircraft parts.
- Conducted BOM inspection over **60+** engineering drawings for pressurized hoses, for Parker Hannifin and Bell Flight, correcting **173 (20% of total)** critical discrepancies in **Excel** improving inventory control and mitigating assembly errors.
- Created **Solidworks** assembly model of hose pressure testing manifold with **~80** parts per assembly with two configuration options depending on manufacturing preference increasing testing capacity by **8x**.
- Reverse engineering of Orbitread hub, reducing weight by **12%** in a 49 lb. assembly by changing material from steel to aluminum, decreasing risk of injury for our technicians working with possible lifting cycles of 80x per day.

PROJECTS

FSAE Beam Wing Oct 2025 - Dec 2025

- Achieved net ~5 lb increase in downforce between beam wing and undertray improvements and rearward center-of-pressure shift; identified wake interference as a limiting factor
- Researched NACA airfoil geometries to design a rear beam wing for CPP FSAE maintaining a 6-week R&D timeline
- Conducted **21+ CFD** simulations with **SolidWorks** and in **ANSYS Fluent** to evaluate downforce, wake behavior, and aerodynamic efficiency with drag considering other aerodynamic tools
- Determined addition of a beam wing would be incoherent with the overall goals and cost of the current aero package

CNC Router Apr 2026-May 2026

- Built an autonomous CNC drill prototype using **SolidWorks**, **Arduino**, NEMA 17 stepper motors, belts, lead screws, and aluminum extrusions under a $200 budget to drill required hole locations, depths, and diameters into a foam block..
- Developed code in Arduino to smooth out stepper motor backlash improving accuracy by **74%**, without increasing time drilling all holes in approximately 1 minute and 40 seconds.
- Developed a belt driven linear actuator system to move the drill in X and Y with interference fit bearings designed to glide along aluminum extrusions for reliable and consistent movements.
- Created CAD assemblies and engineering drawings for the carriage, block holder, Z-axis frame, and spindle components.

Knife Design and Fabrication March 2025 - June 2025

- Engineered a fixed-blade knife using A2 tool steel, heat treatment to achieve a post-temper hardness of 51.5 HRC and maximum bending load of 13.65 KN using **Universal Instron**
- Modeled with high-saber grind and custom PLA Wa-style handle, friction fit sheath using **SolidWorks** and **3D printing**
- Conducted full fabrication cycle including beveling, tempering, and sharpening using hand tools and wet stone from McMaster-Carr stock material, achieving the fastest slicing time (0.79 s) in the final evaluation.

Winch Sled Jan 2026 - Mar 2026

- Designed a motorized transport system to move four 500-g masses across a 5-ft distance and onto a 30-in raised platform using one 6-V DC motor, three simple machines, and a sub-$100 budget.
- Performed SolidWorks FEA on the base, pulley supports, and spool under worst-case loading conditions, achieving factors of safety of approximately 13 for the base and 3 for the spool with less than 0.1 mm predicted displacement.
- Collaborated with a four-person team to modify the cart during testing, increasing its capacity from two to three or four masses per cycle and achieving a delivery rate of approximately 4,000 g/min.

Project Car Cost Program Oct 2025 - Dec 2025

- Developed **Python** script program in **Visual Studio Code** to organize 38 data points and $16,000 worth of cost
- Clean UI interface with tkinter and DB browser for data management for ease of use limiting complexity
- Organizes tabs for 14 customers at automechanic shop improving transparency and confidence